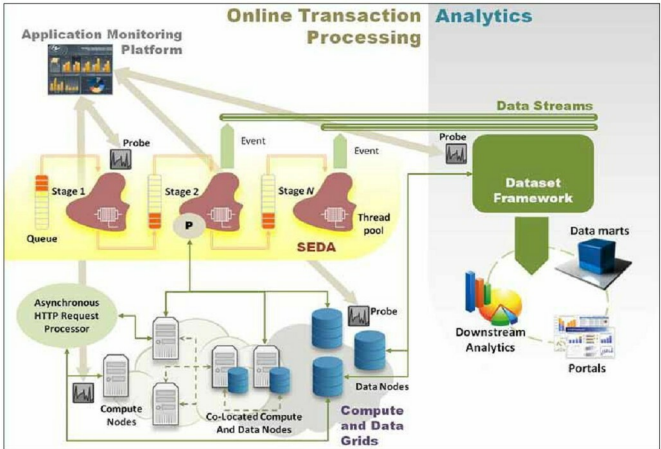




Big data platform used to implement Aadhaar

But have there been any pitfalls during FOSS adoption? To this question, he replies: "The challenges to using OSS are that of perception, the willingness of developers to explore and the availability of support for all the solutions available out there."

FOSS tools that are powering the UIDAI project

For the Aadhaar project, the UIDAI mainly uses FOSS solutions, a list of which is given below.

RabbitMQ: This is an AMQP (advanced message queuing protocol) standards-based messaging solution for many of the project's asynchronous processing needs, with integration and high scalability via the SEDA (staged event-driven architecture) implemented in Mule.

Mule: This is a light-weight object broker and SEDA runtime instance used also in orchestrating various stages in batch processing.

The Hadoop stack: HDFS (Hadoop distributed file system) is used to provide high data read/write throughput in the order of many tera-bytes per day. Distributed architecture enables scale out as needed. Hive is used for building the UIDAI data warehouse, HBase for indexed lookup of records across millions of rows, Zookeeper as a distributed coordination service for server instances, and Pig as an ETL (extract, transform and load) solution for loading data into Hive.

Pentaho is used for report generation and OLAP (on-line

The FOSS advantages

From his own experience with the open source software stack, Balasubramanian enumerates the key benefits that the use and adoption of OSS can render to developers, software product development teams and organisations, respectively.

"For developers: OSS encourages getting your hands dirty and promotes the geek in you. OSS users are therefore better off than their proprietary software counterparts in their computer science skills. Access to the source of well-written frameworks enhances one's own design and programming skills. The can-do attitude of OSS users and contributors makes them better suited to solve complex design and programming issues.

"For software product teams: Many OSS frameworks have liberal licensing and distribution terms. This allows product teams to embed them into their proprietary solutions. Product teams that open source their solutions can still benefit from business models of support, multiple distributions, training, etc.

"Organisations can also save on licence costs by using low-risk, well-written tools like source code editors and such. Organisations that have a strong engineering culture and development teams to support their IT solutions are better suited to adopt OSS. Such organisations typically have OSS solutions in their roadmap and influence business units to build solutions on it. In general, overall costs can be low when adopting OSS that is stable and has an active user community. Finding skilled resources for mainstream OSS solutions is not difficult or expensive."